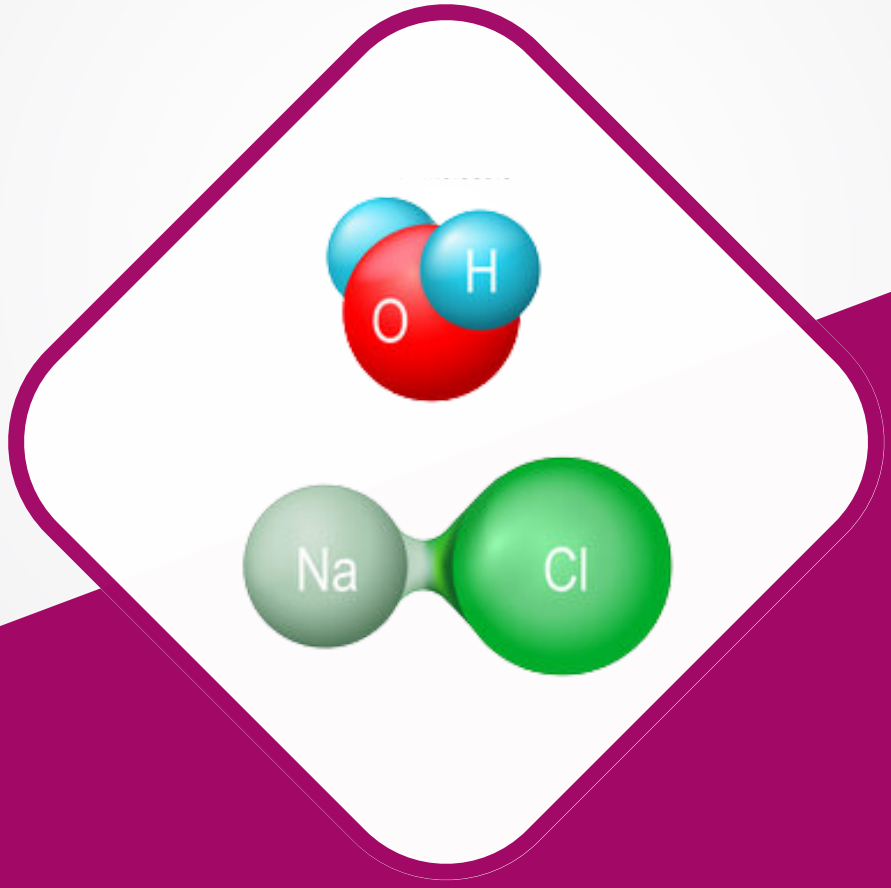
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Inorganic Chemistry

ENTHUSIAST COURSE



STUDY MATERIAL

Reactions of Salt Analysis

English Medium

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Tabulating Reactions of Group-I Cation

S.No.	Reagents	Pb^{+2}	Hg_2^{+2}	Ag^+
1.	(a) Group reagent HCl	$PbCl_2$ (White)	Hg_2Cl_2 (White)	AgCl (White)
	(b) HCl + NH_3	$PbCl_2$ + $Pb(OH)_2$ (White ppt.)	(Black) Hg + $HgNH_2Cl$ (White)	Dissolve $[Ag(NH_3)_2]^+$
	(c) HCl + boiling	Dissolves	No change	No change
2.	(a) KI, small amount	(Yellow) PbI_2	Hg_2I_2 (Green)	AgI (Yellow)
	(b) KI Excess	No change Conc. I^-(6M) + $PbI_2 \rightarrow [PbI_4]^{-2}$	Grey Hg + $[HgI_4]^{-2}$	No change
3.	K_2CrO_4	(Yellow) $PbCrO_4$	(Red) Hg_2CrO_4	Brick red Ag_2CrO_4
4.	(a) KCN	(White) $Pb(CN)_2$	(Black) Hg + $Hg(CN)_2$ soluble	(White) AgCN
	(b) Excess KCN	No change	No Change	$[Ag(CN)_2]^-$ soluble
5.	(a) NaOH small amount	$Pb(OH)_2$ (White)	Hg_2O(Black)	Ag_2O (Brown)
	(b) Excess of NaOH	Dissolves $[Pb(OH)_4]^{-2}$	No change	No change
6.	(a) NH_3 small amount	$Pb(OH)_2$ (White)	Hg(black)+$HgO.HgNH_2NO_3$(White)	Ag_2O (Brown)
	(b) Excess NH_3	No Change	No Change	$[Ag(NH_3)_2]^+$ soluble
7.	H_2S in acidic medium (H_2S + HCl)	PbS (Black)	Black Hg + HgS	Black Ag_2S
8.	(a) Na_2CO_3	White ppt of a mixture of $Pb(OH)_2$+$PbCO_3$ or $PbO.PbCO_3$	Hg_2CO_3 yellowish white	Ag_2CO_3 yellowish white
	(b) Na_2CO_3 + Boiling	No change	$\underbrace{\text{(Black) Hg + HgO} \downarrow \text{ (Yellow)}}_{\text{Blackish–grey}}$	Brown Ag_2O

9.	Na_2HPO_4	(White) $Pb_3(PO_4)_2$	(White) Hg_2HPO_4	(Yellow) Ag_3PO_4
10.	(a) Na_2SO_3	$PbSO_3$ (White)	–	No visible change occurs due to formation of complex initially but the addition of more reagent white Ag_2SO_3 is formed
	(b) Boiling	ppt. oxidized by atmospheric oxygen and $PbSO_4$ is formed	–	Grey metallic silver is precipitated
11.	(a) $Na_2S_2O_3$	First there is no change but further addition of the reagent a white ppt. of PbS_2O_3 is formed	–	At first no ppt occurs because the soluble complex $[Ag(S_2O_3)_2]^{-3}$ is formed but further addition of the reagent a white ppt of $Ag_2S_2O_3$ is formed
	(b) Boiling	On boiling the suspension the precipitate darkens forming finally black ppt of PbS ↓	–	The ppt. is unstable, turning dark on standing, when Ag_2S is formed this hydrolytic decomposition can be accelerated by warming
12.	Specific reactions	(i) With K_2CrO_4 ↓ yellow $PbCrO_4$ ↓ (ii) With Diphenylthiocarbazone gives brick red complex ↓	With Di phenyl carbazide ↓ (violet colour)	With p-dimethylamino benzylidenerhodanine ($+HNO_3$) → (violet colour).

Tabulating Reactions of Group-II Cation

S.No.	Reagents	Pb^{+2}	Cu^{+2}	Hg^{+2}	Cd^{+2}	Bi^{+3}
1.	Group reagent H_2S gas in acidic medium	PbS(Black)$\downarrow$	CuS (Black) $\downarrow$	HgS (Black) $\downarrow$	CdS (Yellow) $\downarrow$	Bi_2S_3(Brownish Black)$\downarrow$
2.	(a) KI (Little)	$PbI_2\downarrow$(deep yellow)	$\underbrace{CuI(White)\downarrow + I_3^-(Brown)}_{\text{Brown colouration}}$	HgI_2 (Scarlet Red)	No reaction	BiI_3(Black)$\downarrow$
	(b) Excess KI	$PbI_{2(s)} + I^-(6M) \rightarrow [PbI_4]^{-2}$ soluble	No reaction	$[HgI_4]^{-2}$ Soluble (colourless)	–	$BiI_3 + I^- \rightleftharpoons [BiI_4]^-$ Orange solution when diluted with water the above reaction is reversed and black BiI_3 is reprecipitated $BiI_3 \xrightarrow[H_2O]{\text{Boil}} BiOI$ (orange turbidity)
3.	(a) KCN (Little)	$Pb(CN)_2$(white ppt.)$\downarrow$	$Cu(CN)_2$ (Yellow) $\rightarrow$ $CuCN\downarrow$(White)+$(CN)_2\uparrow$	$Hg(CN)_2$(soluble)	$Cd(CN)_2$ (White ppt.)	$Bi(CN)_3$ (White ppt.)
	(b) Excess KCN	No reaction	$K_3[Cu(CN)_4]$ colourless solution	–	$K_2[Cd(CN)_4]$ colourless solution	No reaction
4.	KSCN	$Pb(SCN)_2\downarrow$ (White)	$Cu(SCN)_2\downarrow$ (Black) $\rightarrow$ $CuSCN\downarrow + (SCN)_2$ (White)	$Co^{+2}+4SCN^-+Hg^{+2} \rightarrow Co[Hg(SCN)_4]\downarrow$ (Deep Blue ppt.)	No reaction	$Bi(SCN)_3$ (White)
5.	(a) NaOH	$Pb(OH)_2$ (White ppt.)$\downarrow$	$Cu(OH)_2$ (Light blue ppt.$\downarrow$)	HgO (Yellow ppt.)	$Cd(OH)_2$ (White ppt.)	$Bi(OH)_3$ (White ppt.)
	(b) Excess NaOH	$[Pb(OH)_4]^{-2}$ soluble	Insoluble (No change)	Insoluble (No change)	Insoluble (No change)	Insoluble (No change)
6.	$SnCl_2$	$PbCl_2$ (White)	$CuCl$ (White)	Hg_2Cl_2 (White) if excess $SnCl_2$ then Hg (grey-black)	(No reaction)	Bi (Black) in basic medium

Tabulating Reactions of Group-III Cation

S.No.	Reagents	Al^{+3}	Cr^{+3}	Fe^{+3}	Fe^{+2}
1.	KI	–	–	$Fe^{+3}+I^- \rightarrow Fe^{+2} + I_2$ (redox)	$Fe^{+2} + I^- \rightarrow FeI_2$
2.	(i) KCN	–	–	$Fe(CN)_3$ (Brown ppt.)	$Fe(CN)_2$ (Yellowish–brown)
	(ii) Excess KCN	–	–	$K_3[Fe(CN)_6]$ (Yellow solution)	$K_4[Fe(CN)_6]$ (Pale-yellow sol.)
3.	KSCN	–	–	$[Fe(H_2O)_5(SCN)]^{+2}$ Blood red colouration	No colouration
4.	(a) $(NH_4)_2S$	$Al(OH)_3$ (White gelatinous ppt.)	$Cr(OH)_3$ (Green ppt.)	$FeS\downarrow$ (Black) + $S\downarrow$ (White) → Black ppt.	FeS(Black)
	(b) If add HCl	$Al(OH)_3$ dissolve	$Cr(OH)_3$ dissolve	Turns white because FeS is dissolve $FeCl_2 + H_2S\uparrow + S$(White ppt.)	FeS dissolves $FeCl_2 + H_2S\uparrow$
5.	(a) NaOH (Little)	$Al(OH)_3$ (White)	$Cr(OH)_3$ (Green)	$Fe(OH)_3$ (Reddish-brown)	$Fe(OH)_2$ (Dirty Green)
	(b) Excess NaOH	Dissolves $[Al(OH)_4]^-$	Dissolve$[Cr(OH)_4]^-$(Green)	Insoluble	Insoluble
6.	(a) NH_3(Little)	$Al(OH)_3$ (White)	$Cr(OH)_3$ (Green ppt.)	$Fe(OH)_3$ (Reddish-brown)	$Fe(OH)_2$ (Dirty Green)
	(b) Excess NH_3	Insoluble	Slightly soluble due to formation of $[Cr(NH_3)_6]^{+3}$ violet or pink solution	No reaction	No reaction

7.	$K_4[Fe(CN)_6]$	–	–	$Fe_4[Fe(CN)_6]_3$ (Prussian Blue) soluble in oxalic acid	$K_2Fe[Fe(CN)_6]$ (White)
8.	$K_3[Fe(CN)_6]$	–	–	$Fe[Fe(CN)_6]$ (Brown)	$Fe_3[Fe(CN)_6]_2$ (Turnbull's Blue)
9.	Specific test	Add CH_3COONa no change but boil $Al(OH)_2CH_3COO\downarrow$ (White ppt.)	Add 1,5-Dipehnylcarbazide ↓ (Violet colour) (a) With $Na_2O_2 \rightarrow$ CrO_4^{2-} (Yellow) (b) With $BO_3^- + OH^- \rightarrow$ CrO_4^{2-} (Yellow) (c) With $OBr^- \rightarrow$ CrO_4^{2-} (Yellow) (d) With $S_2O_8^{2-} \rightarrow$ CrO_4^{2-} (Yellow)	Add $CH_3COONa \rightarrow$ Deep red colour complex $[Fe(OH)_2(CH_3COO)_6]^+$ On boiling $Fe(OH)_2CH_3COO$ (Brownish Red ppt.)	Add dipy or phen. red colour Add dmg (ammonical) Red colouration (Ni^{+2}, Co^{+2}) interfere
10.	Na_2CO_3	$Al(OH)_3$ (White)	$Cr(OH)_3$ (Green)	$Fe(OH)_3$ (Reddish-brown)	–

Tabulating Reactions of Group-IV Cation

S.No.		Reagents	Ni^{+2}	Co^{+2}	Mn^{+2}	Zn^{+2}
1.	(i)	$(NH_4)_2S$	NiS (Black)	CoS (Black)	MnS (Pink or Buff coloured)	ZnS (White)
	(ii)	Add HCl in sulphide ppt.	No change (requires aqua regia)	No Change (requires aqua regia)	Dissolves, $Mn^{+2} + H_2S \uparrow$	dissolves $Zn^{+2} + H_2S \uparrow$
2.		H_2S (No reaction)	NiS (Partial ppt.)	No reaction	No reaction	Partial precipitation of Zns in neutral solution. When conc. of acid (HCl) is about 0.3 M [pH about 0.6], no ppt.
3.	(i)	Aq. NaOH (Little)	$Ni(OH)_2$ (Green)	Co(OH)Cl (Blue)	$Mn(OH)_2$ (Pink ppt.)	$Zn(OH)_2$ (White)
	(ii)	Excess NaOH	Insoluble	$Co(OH)_2$ (Pink ppt.)	Insoluble	$[Zn(OH)_4]^{-2}$ or Na_2ZnO_2 Soluble colourless
4.	(i)	NH_3 (Little)	$Ni(OH)_2$ (Green)	Co(OH)Cl (Blue)	$Mn(OH)_2$ (Pink ppt.)	$Zn(OH)_2$ (White)
	(ii)	Excess NH_3	$[Ni(NH_3)_4]^{+2}$ (Blue solution)	$[Co(NH_3)_6]^{+2}$ (Yellow)	Insoluble	$[Zn(NH_3)_4]^{+2}$ Colourless solution
5.		KCN (Little)	$Ni(CN)_2$ (Green ppt.)	$Co(CN)_2$ (Reddish-brown)	$Mn(CN)_2.Mn(OH)_2$ Buff	$Zn(CN)_2$ (White)
6.		Excess KCN	$[Ni(CN)_4]^{-2}$ (Soluble)	$[Co(CN)_6]^{4-}$ (Brown solution)	–	–
7.		$K_4[Fe(CN)_6]$	$Ni_2[Fe(CN)_6]\downarrow$ (Light Green)	$Co_2[Fe(CN)_6]$ (Green)	$Mn_2[Fe(CN)_6]$ (White)	$K_2Zn_3[Fe(CN)_6]_2$ (White ppt.)
8.		Specific Test	Add. dmg (ammonical) Pink / Rosy Red ppt. of $[Ni(dmg)_2]$ similar colouration produce by Fe^{+2},Co^{+2} with dmg^{-1}	Add. KNO_2 + $CH_3COOH + CO_2 \rightarrow$ $K_3[Co(NO_2)_6]+NO\uparrow+H_2O$ Yellow ppt. (Fisher salt) $Co^{+2}+4SCN^{-}\rightarrow$ $[Co(SCN)_4]^{-2}$ (Blue solution)	– –	– –

Tabulating Reactions of Group-V Cation

S.No.	Reagents	Ba^{+2}	Sr^{+2}	Ca^{+2}
1.	$(NH_4)_2CO_3$ in presence of NH_4Cl and NH_4OH	$BaCO_3\downarrow$(White)	$SrCO_3\downarrow$ (White)	$CaCO_3\downarrow$(White)
2.	CH_3COOH	$BaCO_3$ is soluble	$SrCO_3$ is soluble	$CaCO_3$ is soluble
3.	$(NH_4)_2CrO_4$	$BaCrO_4$ (Yellow)	$SrCrO_4$ yellow	$CaCrO_4$ (Yellow) but with conc. $(NH_4)_2$ CrO_4
4.	CH_3COOH	$BaCrO_4$ insoluble in dil. CH_3COOH	$SrCrO_4$ soluble in CH_3COOH	$CaCrO_4$ soluble in CH_3COOH
5.	$(NH_4)_2SO_4$	$BaSO_4$(White)	$SrSO_4$ (White)	$CaSO_4$ (White)
6.	Excess of $(NH_4)_2SO_4$	$BaSO_4$ is insoluble	$SrSO_4$ insoluble	$CaSO_4$ is soluble due to formation of complex $[Ca(SO_4)_2]^{-2}$
7.	$(NH_4)_2C_2O_4$	BaC_2O_4 (White) readily dissolved by hot dil. CH_3COOH	SrC_2O_4 (White) CH_3COOH does not attack it so insoluble	CaC_2O_4 (White) insoluble in CH_3COOH

Test of Cations With Na_2HPO_4 Reagent

HPO_4^{-2} →

- Pb^{+2} → $Pb_3(PO_4)_2$ (White ppt.)
- Ag^{+} → Ag_3PO_4 (Yellow ppt.)
- Hg_2^{+2} → Hg_2HPO_4 (White ppt.)
- Fe^{+3} → $FePO_4$ (Yellowish white ppt.)
- Ba^{+2} → $BaHPO_4$ [If NH_4Cl present then $Ba_3(PO_4)_2$ (White ppt.)]
- Al^{+3} → $AlPO_4$ (White ppt.)
- Zn^{+2}
  - If NH_4Cl is present → $Zn(NH_4)PO_4$ (White)
  - If NH_4Cl is absent → $Zn_3(PO_4)_2$ (White ppt.)
- Mn^{+2}
  - If NH_4Cl is present → $Mn(NH_4)PO_4$ (Pink ppt.)
  - If NH_4Cl is absent → $Mn_3(PO_4)_2$ (Pink ppt.)
- Mg^{+2} (in NH_4Cl / NH_4OH) → $Mg(NH_4)PO_4$ (White ppt.) $\xrightarrow{\Delta}$ $Mg_2P_2O_7$ (White ppt.)

Test of Cations With $K_4[Fe(CN)_6]$ Reagent

$K_4[Fe(CN)_6]$ →

- Fe^{+2} → $K_2Fe[Fe(CN)_6]$ (White ppt.) is gradually oxidized to prussian blue by atomspheric oxygen
- Fe^{+3} → $Fe_4[Fe(CN)_6]_3$ (Prussian blue ppt.) is decomposed by solutions of alkalihydroxide, brown iron(III) hydroxide being formed
- Cu^{+2} → $Cu_2[Fe(CN)_6]$ (Chocolate-Brown ppt.) Insoluble in acetic acid but decomposes in solution of alkalydroxide
- Ag^{+} → $Ag_4[Fe(CN)_6]$ (White ppt.) Insoluble in NH_3 solution and HNO_3 but soluble in KCN and $Na_2S_2O_3$ due to formation of soluble complex, upon warming with conc. HNO_3 the ppt. is converted to orange-red, $Ag_3[Fe(CN)_6] \downarrow$ soluble in NH_3 solution [distinction from hexacyanoferrate(II)]
- Cd^{+2} → $Cd_2[Fe(CN)_6]$ (Bluish-White ppt.)
- Ca^{+2} → $K_2Ca[Fe(CN)_6]$ (White ppt.)
- Zn^{+2} → $K_2Zn_3[Fe(CN)_6]_2$ (White ppt.)
- Mn^{+2} → $Mn_2[Fe(CN)_6]$ (White ppt.)
- Co^{+2} → $Co_2[Fe(CN)_6]$ (Green ppt.)
- Ni^{+2} → $Ni_2[Fe(CN)_6]$ (Light green ppt.)